

Commerce XI Maths App

Exercise 1.1 - Detailed Answers with Explanations

Total questions: 34 | +2 correct | -1 wrong

This answer guide follows the question order used in the app.

Question 1 - Q1(i)

Describe the following sets in Roster form
 $\{x/x \text{ is a letter of the word 'MARRIAGE'}\}$

Options

- A. $\{M, A, R, I, G, E\}$
- B. $\{M, A, R, I, G\}$
- C. $\{M, A, R, I, E\}$
- D. $\{M, A, R, I, G, E, S\}$

Correct answer: $\{M, A, R, I, G, E\}$

Explanation and steps

1. The governing definition or property gives: $\{M, A, R, I, G, E\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{M, A, R, I, G, E\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 2 - Q1(ii)

Describe the following sets in Roster form
 $\{x/x \text{ is an integer, } -1/2 < x < 9/2\}$

Options

- A. $\{0, 1, 2, 3, 4\}$
- B. $\{-1, 0, 1, 2, 3, 4\}$
- C. $\{0, 1, 2, 3, 4, 5\}$
- D. $\{1, 2, 3, 4\}$

Correct answer: $\{0, 1, 2, 3, 4\}$

Explanation and steps

1. The governing definition or property gives: $\{0, 1, 2, 3, 4\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{0, 1, 2, 3, 4\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 3 - Q1(iii)

Describe the following sets in Roster form
 $\{x/x = 2n, n \in \mathbb{N}\}$

Options

- A. $\{2, 4, 6, 8, \dots\}$
- B. $\{0, 2, 4, 6, \dots\}$
- C. $\{1, 3, 5, 7, \dots\}$
- D. $\{2, 4, 6, 8\}$

Correct answer: $\{2, 4, 6, 8, \dots\}$

Explanation and steps

1. The governing definition or property gives: $\{2, 4, 6, 8, \dots\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{2, 4, 6, 8, \dots\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 4 - Q2(i)

Which set-builder form represents $\{0\}$?

(W is the set of whole numbers and N is the set of natural numbers.)

Options

- A. $\{x / x \in W, x \notin N\}$
- B. $\{x \mid x \in W, x = 1\}$
- C. $\{x \mid x \in W, x < 2\}$
- D. $\{x \mid x \in Z, x < 1\}$

Correct answer: $\{x / x \in W, x \notin N\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x \in W, x \notin N\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x \in W, x \notin N\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 5 - Q2(ii)

Which set-builder form represents $\{0, \pm 1, \pm 2, \pm 3\}$?

Options

- A. $\{x / -3 \leq x \leq 3, x \in Z\}$
- B. $\{x \mid x \in Z, -3 < x < 3\}$

C. $\{x \mid x \in \mathbb{N}, -3 \leq x \leq 3\}$

D. $\{x \mid x \in \mathbb{Z}, -3 \leq x < 3\}$

Correct answer: $\{x \mid -3 \leq x \leq 3, x \in \mathbb{Z}\}$

Explanation and steps

1. The governing definition or property gives: $\{x \mid -3 \leq x \leq 3, x \in \mathbb{Z}\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x \mid -3 \leq x \leq 3, x \in \mathbb{Z}\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 6 - Q2(iii)

Which set-builder form represents $\{1/2, 2/5, 3/10, 4/17, 5/26, 6/37, 7/50\}$?

Options

A. $\{x \mid x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$

B. $\{x \mid x = n/(n^2 + 1), n \in \mathbb{N}, n < 7\}$

C. $\{x \mid x = n/(n + 1), n \in \mathbb{N}, n \leq 7\}$

D. $\{x \mid x = n^2/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$

Correct answer: $\{x \mid x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$

Explanation and steps

1. The governing definition or property gives: $\{x \mid x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x \mid x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 7 - Q3(i)

Let $A = \{x \mid 6x^2 + x - 15 = 0\}$, $B = \{x \mid 2x^2 - 5x - 3 = 0\}$, and $C = \{x \mid 2x^2 - x - 3 = 0\}$.

Find $A \cup B \cup C$.

Options

A. $\{-5/3, -1, -1/2, 3/2, 3\}$

B. $\{-5/3, 3/2, 3\}$

C. $\{-5/3, -1/2, 3/2, 3\}$

D. $\{-5/3, -1, -1/2, 3/2\}$

Correct answer: $\{-5/3, -1, -1/2, 3/2, 3\}$

Explanation and steps

1. The governing definition or property gives: $\{-5/3, -1, -1/2, 3/2, 3\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{-5/3, -1, -1/2, 3/2, 3\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 8 - Q3(ii)

Let $A = \{x \mid 6x^2 + x - 15 = 0\}$, $B = \{x \mid 2x^2 - 5x - 3 = 0\}$, and $C = \{x \mid 2x^2 - x - 3 = 0\}$.
Find $A \cap B \cap C$.

Options

- A. $\{ \}$
- B. $\{3/2\}$
- C. $\{3\}$
- D. $\{-1/2\}$

Correct answer: $\{ \}$

Explanation and steps

1. The governing definition or property gives: $\{ \}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{ \}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 9 - Q4

Let A, B, and C be the sets of distinct letters in “college”, “marriage”, and “luggage”.
What is $A - (B \cup C)$?

Options

- A. $\{c, o\}$
- B. $\{o\}$
- C. $\{c, o, l\}$
- D. $\{c, o, g\}$

Correct answer: $\{c, o\}$

Explanation and steps

1. The governing definition or property gives: $\{c, o\}$.
2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $\{c, o\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 10 - Q5(i)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Evaluate $A \cup (B \cap C)$.

Options

A. $\{1, 2, 3, 4, 5, 6\}$

B. $\{3, 4, 5, 6\}$

C. $\{1, 2, 3, 4, 5, 6, 7, 8\}$

D. $\{4, 5, 6\}$

Correct answer: $\{1, 2, 3, 4, 5, 6\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 2, 3, 4, 5, 6\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 2, 3, 4, 5, 6\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 11 - Q5(ii)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Evaluate $A \cap (B \cup C)$.

Options

A. $\{3, 4\}$

B. $\{4\}$

C. $\{3, 4, 5, 6\}$

D. $\{1, 2, 3, 4\}$

Correct answer: $\{3, 4\}$

Explanation and steps

1. The governing definition or property gives: $\{3, 4\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{3, 4\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 12 - Q5(iii)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Evaluate $(A \cup B)'$.

Options

- A. $\{7, 8, 9, 10\}$
- B. $\{1, 2, 3, 4, 5, 6\}$
- C. $\{5, 6, 7, 8, 9, 10\}$
- D. $\{7, 8\}$

Correct answer: $\{7, 8, 9, 10\}$

Explanation and steps

1. The governing definition or property gives: $\{7, 8, 9, 10\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{7, 8, 9, 10\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 13 - Q5(iv)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Evaluate $(A \cap B)'$.

Options

- A. $\{1, 2, 5, 6, 7, 8, 9, 10\}$
- B. $\{3, 4\}$
- C. $\{1, 2, 5, 6\}$
- D. $\{5, 6, 7, 8, 9, 10\}$

Correct answer: $\{1, 2, 5, 6, 7, 8, 9, 10\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 2, 5, 6, 7, 8, 9, 10\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 2, 5, 6, 7, 8, 9, 10\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 14 - Q5(v)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.
Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
Evaluate $(A \cap B) \cup (A \cap B')$.

Options

- A. $\{1, 2, 3, 4\}$
- B. $\{3, 4\}$
- C. $\{1, 2\}$
- D. $\{1, 2, 3, 4, 5, 6\}$

Correct answer: $\{1, 2, 3, 4\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 2, 3, 4\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 2, 3, 4\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 15 - Q5(vi)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.
Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
Evaluate $(A \cap B) \cup (A' \cap B)$.

Options

- A. $\{3, 4, 5, 6\}$
- B. $\{3, 4\}$
- C. $\{5, 6\}$
- D. $\{1, 2, 3, 4, 5, 6\}$

Correct answer: $\{3, 4, 5, 6\}$

Explanation and steps

1. The governing definition or property gives: $\{3, 4, 5, 6\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{3, 4, 5, 6\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 16 - Q5(vii)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.
Find $n(A \cup B)$.

Options

- A. 6
- B. 8
- C. 4
- D. 2

Correct answer: 6

Explanation and steps

1. The governing definition or property gives: 6.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is 6.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 17 - Q6(i)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.
Find $n(A \cup B)$.

Correct answer: 45

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 45.
3. Therefore, the required answer is 45.

Question 18 - Q6(ii)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.
Find $n(A \cap B)$.

Correct answer: 10

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 10.
3. Therefore, the required answer is 10.

Question 19 - Q6(iii)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.
Find $n(A' \cap B)$.

Correct answer: 10

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 10.
3. Therefore, the required answer is 10.

Question 20 - Q6(iv)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.
Find $n(A \cap B')$.

Correct answer: 25

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 25.
3. Therefore, the required answer is 25.

Question 21 - Q7(i)

Out of 200 students, 35 failed MHT-CET, 40 failed AIEEE, 40 failed IIT, 20 failed MHT-CET and AIEEE, 17 failed AIEEE and IIT, 15 failed MHT-CET and IIT, and 5 failed all three.
How many failed none?

Correct answer: 132

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 132.
3. Therefore, the required answer is 132.

Question 22 - Q7(ii)

Out of 200 students, 35 failed MHT-CET, 40 failed AIEEE, 40 failed IIT, 20 failed MHT-CET and AIEEE, 17 failed AIEEE and IIT, 15 failed MHT-CET and IIT, and 5 failed all three.
How many failed AIEEE or IIT?

Correct answer: 63

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.

2. Substituting the given values and simplifying gives 63.
3. Therefore, the required answer is 63.

Question 23 - Q8(i)

Of 2,000 people, 70% read Marathi newspapers, 50% read English newspapers, and 32.5% read both.

How many read at least one?

Correct answer: 1750

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 1750.
3. Therefore, the required answer is 1750.

Question 24 - Q8(ii)

Of 2,000 people, 70% read Marathi newspapers, 50% read English newspapers, and 32.5% read both.

How many read neither?

Correct answer: 250

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 250.
3. Therefore, the required answer is 250.

Question 25 - Q8(iii)

Of 2,000 people, 70% read Marathi newspapers, 50% read English newspapers, and 32.5% read both.

How many read exactly one of the two?

Correct answer: 1100

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 1100.
3. Therefore, the required answer is 1100.

Question 26 - Q9

In a hostel, 25 students take tea, 20 coffee, and 15 milk. Ten take both tea and coffee, eight take both milk and coffee, none take both tea and milk, and everyone takes at least one. How many students are in the hostel?

Correct answer: 42

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 42.
3. Therefore, the required answer is 42.

Question 27 - Q10(i)

Of 260 people with a skin disorder, 150 were exposed to chemical A, 74 to B, and 36 to both. How many were exposed to A but not B?

Correct answer: 114

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 114.
3. Therefore, the required answer is 114.

Question 28 - Q10(ii)

Of 260 people with a skin disorder, 150 were exposed to chemical A, 74 to B, and 36 to both. How many were exposed to B but not A?

Correct answer: 38

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 38.
3. Therefore, the required answer is 38.

Question 29 - Q10(iii)

Of 260 people with a skin disorder, 150 were exposed to chemical A, 74 to B, and 36 to both. How many were exposed to A or B?

Correct answer: 188

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.

2. Substituting the given values and simplifying gives 188.
3. Therefore, the required answer is 188.

Question 30 - Q11

If $A=\{1,2,3\}$, which is the complete power set $P(A)$?

Options

- A. $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$
- B. $\{\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$
- C. $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}\}$
- D. $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2,3\}\}$

Correct answer: $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

Explanation and steps

1. The governing definition or property gives: $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 31 - Q12(i)

Write the interval $[-3, 0]$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$
- B. $\{x \in \mathbb{R} \mid -3 < x < 0\}$
- C. $\{x \in \mathbb{R} \mid -3 \leq x < 0\}$
- D. $\{x \in \mathbb{R} \mid -3 < x \leq 0\}$

Correct answer: $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 32 - Q12(ii)

Write the interval $[6, 12]$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$
- B. $\{x \in \mathbb{R} \mid 6 < x < 12\}$
- C. $\{x \in \mathbb{R} \mid 6 \leq x < 12\}$
- D. $\{x \in \mathbb{R} \mid 6 < x \leq 12\}$

Correct answer: $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 33 - Q12(iii)

Write the interval $(6, 12]$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$
- B. $\{x \in \mathbb{R} \mid 6 \leq x \leq 12\}$
- C. $\{x \in \mathbb{R} \mid 6 < x < 12\}$
- D. $\{x \in \mathbb{R} \mid 6 \leq x < 12\}$

Correct answer: $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 34 - Q12(iv)

Write the interval $[-23, 5)$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, -23 \leq x < 5\}$
- B. $\{x \in \mathbb{R} \mid -23 < x < 5\}$

C. $\{x \in \mathbb{R} \mid -23 \leq x \leq 5\}$

D. $\{x \in \mathbb{R} \mid -23 < x \leq 5\}$

Correct answer: $\{x \mid x \in \mathbb{R}, -23 \leq x < 5\}$

Explanation and steps

1. The governing definition or property gives: $\{x \mid x \in \mathbb{R}, -23 \leq x < 5\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x \mid x \in \mathbb{R}, -23 \leq x < 5\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.